

Sultanate of Oman
Ministry of Health
Royal Hospital



سلطنة عمان
وزارة الصحة
المستشفى السلطاني

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Title: Pathway for Difficult Pediatric IV access during on call time

Pathway For Difficult Pediatric IV access During On Call Time

Risk Factors

1. Chronic patients or prolonged hospital stay > 5 days
2. Known Difficult IV line
3. Age < 1 year
4. Prematurity, recent discharge from NICU

Expected Difficult IV lines

IV line to be Attempted by On call Pediatric registrar
No > 3 times/attempt

- ☐ Involve HD/PICU teams
- ☐ Consider use of Intra-osseous line (IO) for sick patients

Failed attempts

Stable Patients when fluids/Drugs can be changed to oral/intramuscular(IM) routes

- ☐ IM antibiotics should be only used in clinically stable patients
- ☐ Refer to Pediatric IM antibiotic dosing and administration

- ☐ Discuss the line plan with the attending Consultant
- ☐ Review need for long term line if clinically applicable

Stable Patients with no need for HD/PICU admissions Who Need Life Saving Line such as for TPN or other infusions
☐ Peripheral lines will be used if need for central line can be post pond to following day

Emergency Central Lines Needed

Involve General Pediatric Anesthesia
Page floor manager on 5875 before 21:00/
On call register 5227 /OT control 9787 to discuss with on call consultant

Failed attempts

Involve Pediatric Cardiac Anesthesia page 5377

Involve Pediatric surgical team for surgical Cut-Down

Unstable Patients with life threatening features including:
1.Hemo dynamic instability
2.respiratory distress/Failure
3. Need for immediate life saving measures including mechanical ventilation

- ☐ Involve HD/PICU teams
- ☐ Consider early use of Intra-osseous line (IO)
- ☐ Involve on call Consultant for general pediatric

- ☐ Limit number of attempts and spare IV sites based on operator's experience
- ☐ Central lines preferably done during working hours if peripheral lines are enough for patient's management
- ☐ Consultant-Consultant interaction when referring for emergency central lines

ADMINISTRATION OF IM ANTIBIOTICS FOR URGENCY WHEN AN IV-LINE IS NOT AVAILABLE

This table may be used to guide limited antibiotic use during severe infections of pneumonia, sepsis, or meningitis when no IV access for a child or infant is available. It is not for routine use of IM antibiotic injections.

Medication & Presentation	IM dose	Preparation	Administration	Stability	Comments
Benzylpenicillin (Penicillin G) 1 MU (600 mg) Powder for reconstitution	Neonatal dose 1,000,000 units/kg frequency varies as per gestational age and day of life Pediatric dose 100,000 to 300,000 units/kg/day divided Q4H-6H	Add 1.6 mL of sterile water for injection to the vial (1 MU) Concentration is now equal to 300 mg/mL	Measure prescribed dose Intramuscular injection into thickest part of the vastus lateralis in the anterolateral thigh (maximum 0.5 mL per site for neonates)** Doses requiring volumes exceeding the maximum injection site allowance should be split.	Reconstituted solutions must be used immediately	Use IM route only if IV route not available 1 MU vial contains 41.4 mg of sodium Dosage adjustment in renal impairment
CefOTAXIME 500 mg vial 1 gram vial Powder for reconstitution	Neonatal dose 50 mg/kg/dose frequency varies as per gestational age and day of life Pediatric dose 50 mg/kg/dose Q6H-Q12H	500 mg vial Add 1.8 mL of sterile water for injection or lidocaine/lignocaine 1% (10mg/mL) Concentration is now equal to 250 mg/mL 1g vial Add 3.6 mL of sterile water for injection or lidocaine/lignocaine 1% (10mg/mL) Concentration is now equal to 250 mg/mL	Measure prescribed dose IM injection into thickest part of the vastus lateralis in the anterolateral thigh (maximum 0.5 mL per site in neonates)** Doses requiring volumes exceeding the maximum injection site allowance should be split.	If lidocaine (lignocaine) used administer immediately If SWFI is used, solution is stable for 12 hours at room temperature Protect from light	Use IM route only if IV route not available Lidocaine helps to minimize pain at the injection site Never give IV with lignocaine 1% as diluent Dosage adjustment in renal impairment
CefTRIAXONE 1 gram vial Powder for reconstitution	Neonatal dose* PNA < 14 days: 50 mg/kg/dose every 24 hours PNA > 14 days: 100 mg/kg every 24 hours Pediatric dose 100 mg/kg/dose OD Maximum recommended single IM dose is 2 grams Max: 4000 mg per day	Add 2.3mL of sterile water for injection or lidocaine/lignocaine 1% (10mg/mL) Concentration is now equal to 350mg/mL	Administer up via deep IM injection into a large muscle mass e.g. thigh, buttocks Doses requiring volumes exceeding the maximum injection site allowance should be split.	If lidocaine (lignocaine) used administer immediately If SWFI is used, solution is stable for 12 hours at room temperature Protect from light	Use IM route only if IV route not available Lidocaine helps to minimize pain at the injection site Never give IV with lignocaine 1% as diluent

ADMINISTRATION OF IM ANTIBIOTICS FOR ORGANY WHEN AN IV LINE IS NOT AVAILABLE

Points to consider

IM administration

- The IM site for injection is selected after considering the child's age, the volume of drug to be administered, the level of muscle development present and the size of the needle used.
- In general, IM administration of antibiotics in very low birth weight premature neonates is not practical due to small muscle mass, and absorption is unreliable due to hemodynamic instability that is relatively common in this population.
- *Ceftriaxone should NOT be used:
 - in neonates below 41 weeks corrected gestations age.
 - in neonates above 41 weeks corrected gestations age with jaundice, hypoalbuminemia, and acidosis.
- **IM volume recommendations:
 - Infants: 0.5 mL per site
 - Toddlers: 1 mL per site
 - Older children: 2 mL per site (except if using the deltoid muscle, then 1 mL is considered to be the maximum)
- Medication injected into muscle tissues is absorbed less rapidly and takes effect more slowly than medication that is injected intravenously
- Administration rate: no faster than 1 mL per 10 seconds to facilitate absorption and minimize pain is recommended

Site of IM injection

- If more than one dose/site is needed, ensure that injection site is rotated.
- When two IM injections are being administered, give one medication into the right thigh and the other into the left thigh.
- The anterolateral aspect of the thigh, or the vastus lateralis, is the preferred site for infants and toddlers (birth to 2 years) due to the large muscle mass.
- The deltoid in infants is not sufficiently bulky to absorb IM medications adequately. The vastus lateralis muscle avoids the thicker layer of subcutaneous fat on the anterior thigh.
- The deltoid is the preferred site for IM injections in children ages 3 to 18, once adequate muscle mass has developed there, usually around the age of 3 years.
- Never give an IM injection in the buttocks. Using the vastus lateralis muscle avoids the risk of sciatic nerve damage from gluteal injection. Also, the vastus lateralis muscle has a larger muscle mass than the gluteal region and therefore has reduced risk of severe local reactions.

Contraindications to IM

- Active infection, cellulitis or dermatitis at the site of administration
- Known allergy or hypersensitivity to the drug
- Thrombocytopenia
- Coagulation disorders
- Hypovolemic shock: absorption of medication may be hindered owing to compromised vascularity to the muscle.
- Myopathies
- Associated muscular atrophy: leads to delayed drug absorption as well as adds up the risk of neurovascular complications

Ways to minimize discomfort or pain during an IM injection

- Breastfeed the infant at time of injection if appropriate.
- Oral sucrose may be given for relief of distress with parental consent to infants.
- Distraction techniques may be utilized in older children.

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